

Erythropoiesis

Factors Affecting Erythropoiesis

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LEARNING OBJECTIVES

- At the end of the lecture, student should be able to
- Define erythropoiesis
- Describe about the stem cells
- Enlist the site of erythropoiesis
- Illustrate the different stages of erythropoiesis
- Explain the factors affecting erythropoiesis

Haemopoiesis

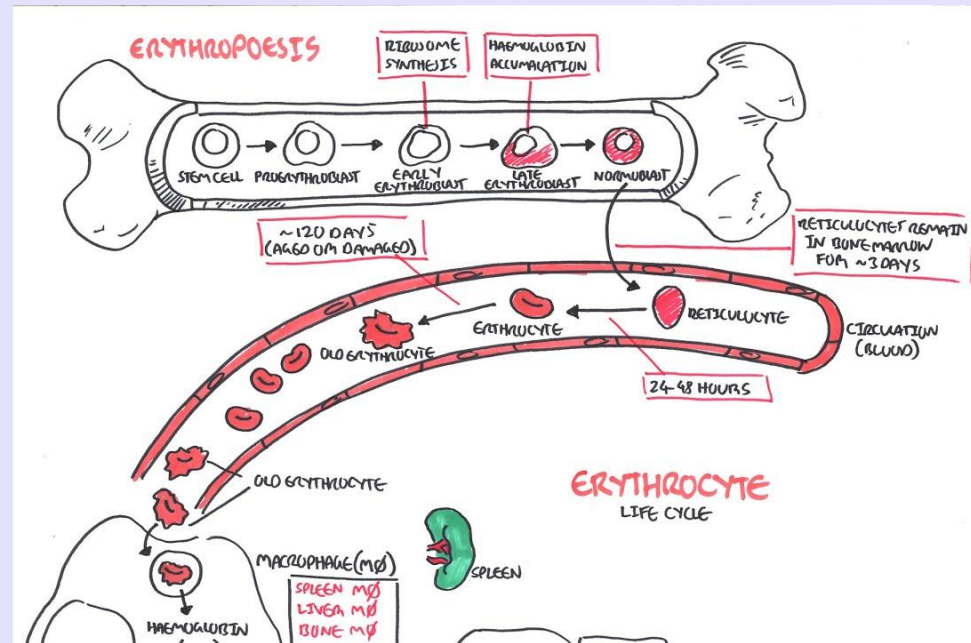
- **Stem cell:** human body has very specialized type of cells called stem cell. They have the remarkable ability to produce many types of other cell of the body.
- When stem cell divides, it has the potential to remain as a stem cell or turn into another type of cell with differentiated and specialized functions.

Haemopoiesis

- **Haemopoiesis** is the process of formation of the formed (solid) elements of blood.
- It starts with the pluripotent stem cells, which are derived from uncommitted totipotent stem cells.
- They have the capability of self renewal.
- There are separate pools of the progenitors cells for **Megakaryocyte**, **Lymphocyte**, **Erythrocytes**, **Eosinophil** and **Basophil**.
- Where as Neutrophil and **monocyte** arise from common precursors

Red Blood Cells

- Total RBCs □ 20 – 30 trillion
- Males □ 5 – 6 million RBC / μL (mm^3)
- Females □ 4 – 5 million RBC / μL (mm^3)
- 1% RBCs destroyed daily & must be replaced
- Some 2-3 million RBCs produced / sec



Erythropoiesis – RBC formation

Origin, development & maturation of red blood cells

Occurs successively in **yolk sac**, **liver**, **spleen** & **BM**

Sites of

Fetus –

3rd week: yolk sac (mesoblastic stage)

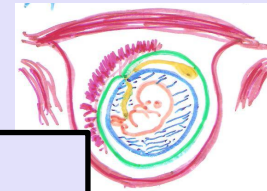
3rd month: liver & spleen (hepatosplenic stage)

Before birth: bone marrow (myeloid stage)

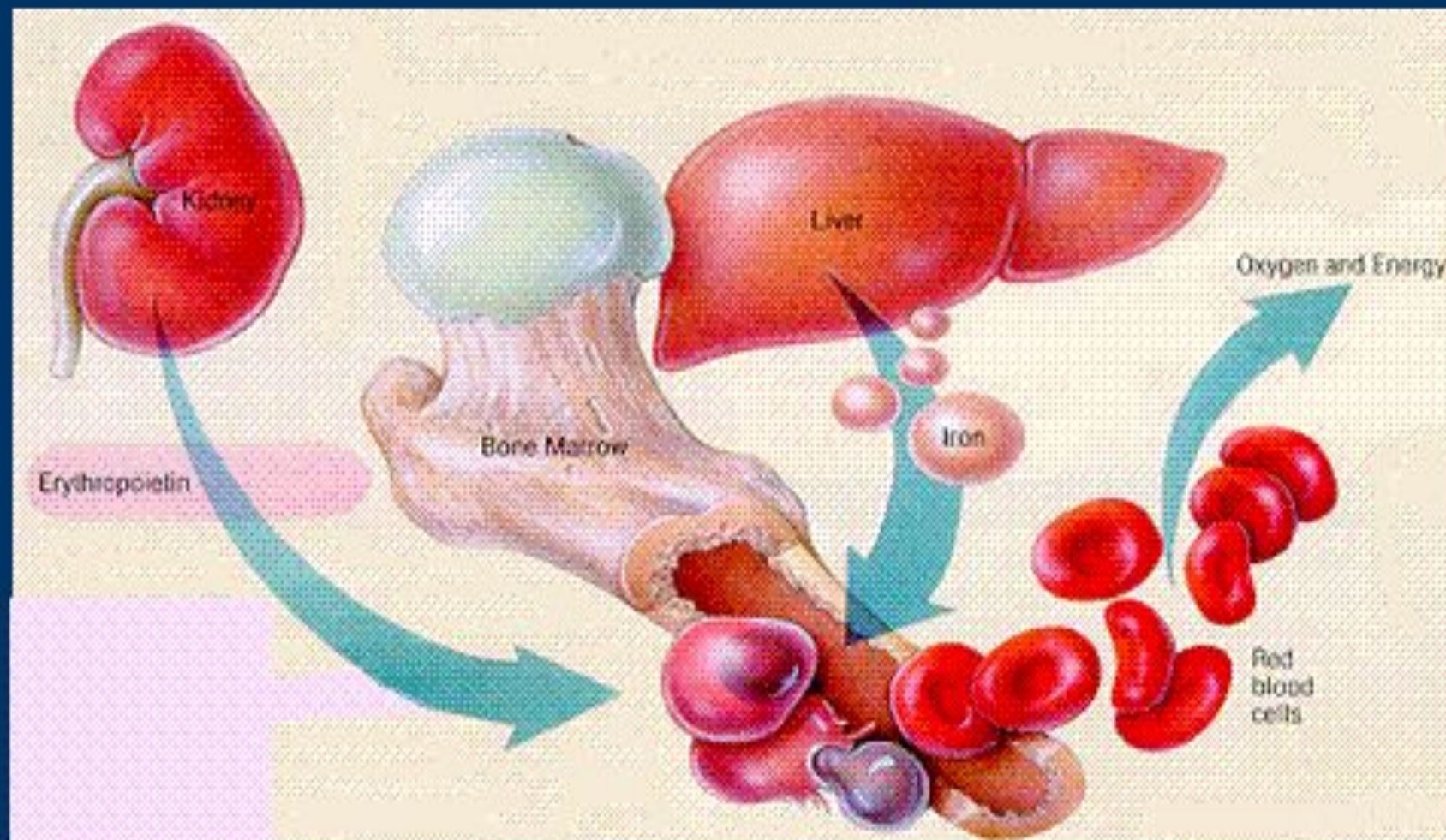
Infant to Puberty – bone marrow of all bones of body

Adult – bone marrow of axial skeleton (skull, Vertebrae, Ribs, Sternum, Sacrum, Pelvis)

bone marrow of prox. ends of humerus &



Normal Erythropoiesis



Growth

Inducers:

- EPO
- CSF
- IL
- Thrombopoietin

Stem cell

Progenitor cell
(CFU)

Precursor cell
(Blast)

(CFU
-E)

HEMOPOIESIS

IL-1
IL-3
IL-6
SCF
G-CSF



Pluripotent
Stem Cell

IL-1
IL-6
SCF
FLT-3L

18-23 um
Large nuc
Thin rim of
cyto

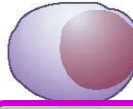
IS



Myeloid
Stem Cell



Lymphoid
Stem Cell



CFU-GM

GM-CSF
IL-3

GM-CSF
IL-3

GM-CSF
IL-3

IL-3



Monoblast

GM-CSF
M-CSF



Myeloblast

GM-CSF
G-CSF



Eosinophilic
Myeloblast

GM-CSF
IL-5



Basophilic
Myeloblast

IL-3
IL-4



Monocyte



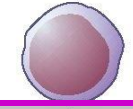
Neutrophil



Eosinophil



Basophil



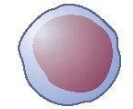
Pre-B Cell

IL-1
IL-2
IL-4



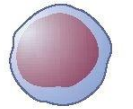
Prothymocyte

IL-2
IL-4



B Lymphoblast

Antigen-
Driven



T Lymphoblast



B Lymphocyte



T Lymphocyte



Proerythroblast

EPO

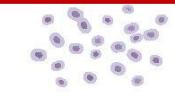


Red Blood Cell



Megakaryocyte

TPO



Platelets

Stem cell: (PHSC) Capable of self renewal; can differentiate into multiple cell types

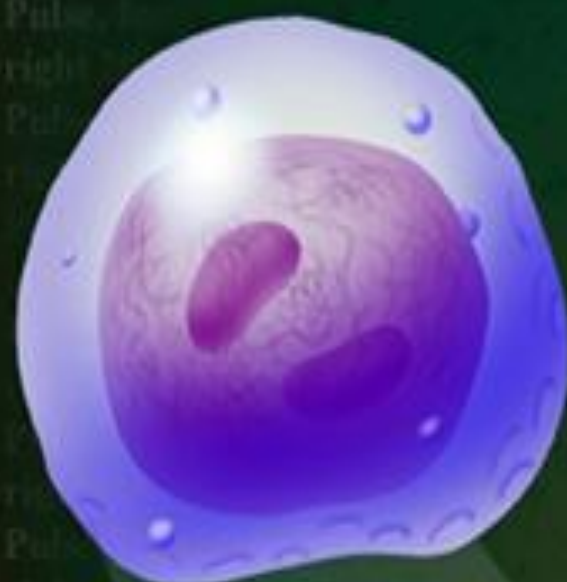
Progenitor cells: (CFU) (Committed) Have a single cell lineage; Proliferate & differentiate into precursor cells in the presence of growth factors; Indistinguishable from stem cells

Precursor cells: (Blasts) Show distinct morphologic characteristics, e.g., Proerythroblast, myeloblast, monoblast

DEFINITION

- **Erythropoiesis** is the process of the origin, development and maturation of erythrocytes. **Hemopoiesis** or hematopoiesis is the process of origin, development and maturation of all the blood cells.
- Erythropoiesis starts from the stem cells in the bone marrow which are pluripotent stem cell.

Erythropoiesis



Proerythroblast



Basophilic erythroblast



Polychromatic erythroblast



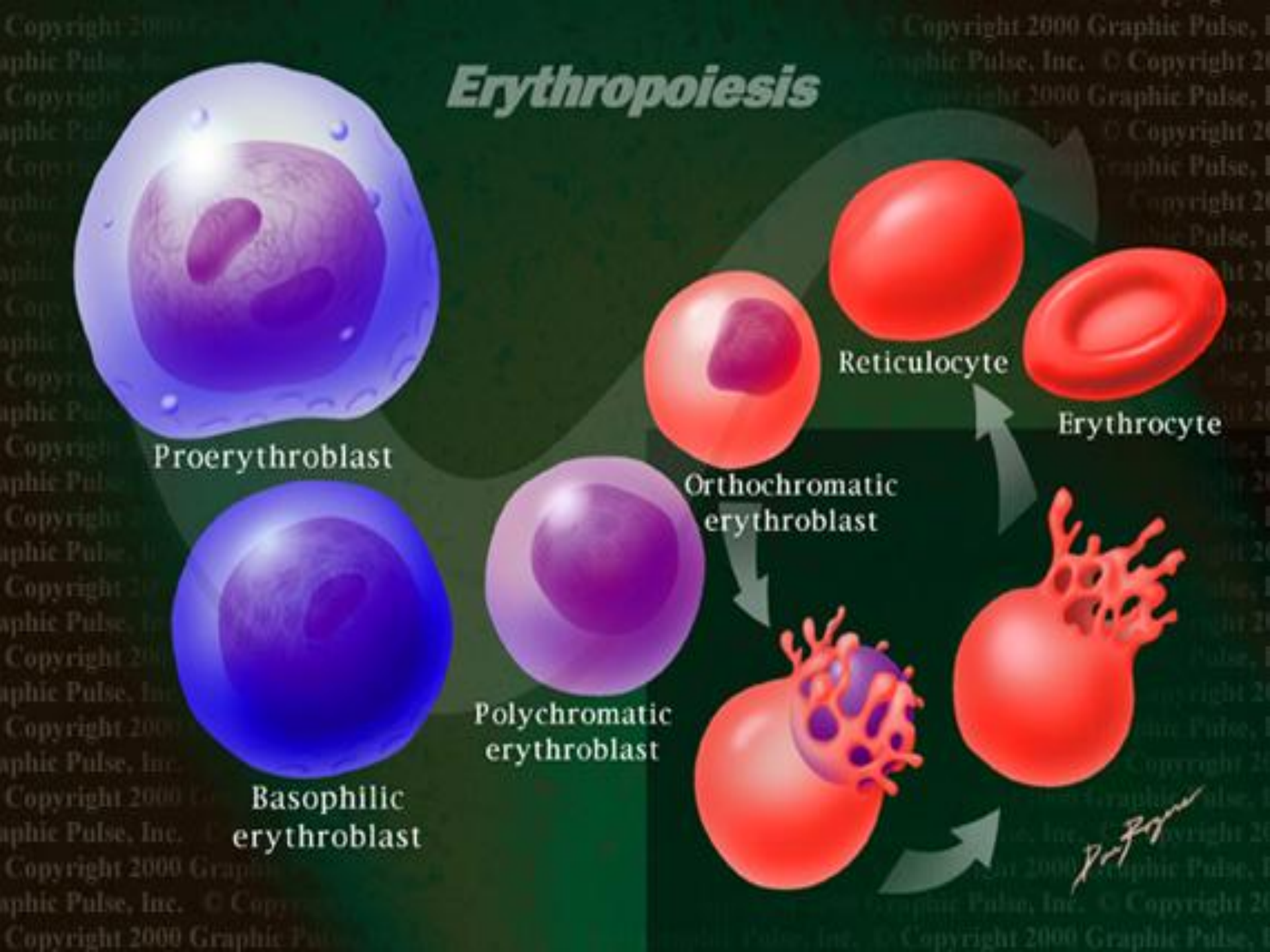
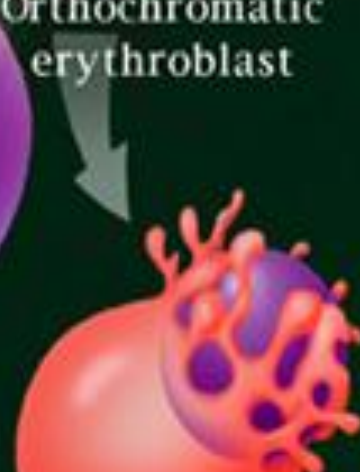
Orthochromatic erythroblast



Reticulocyte



Erythrocyte





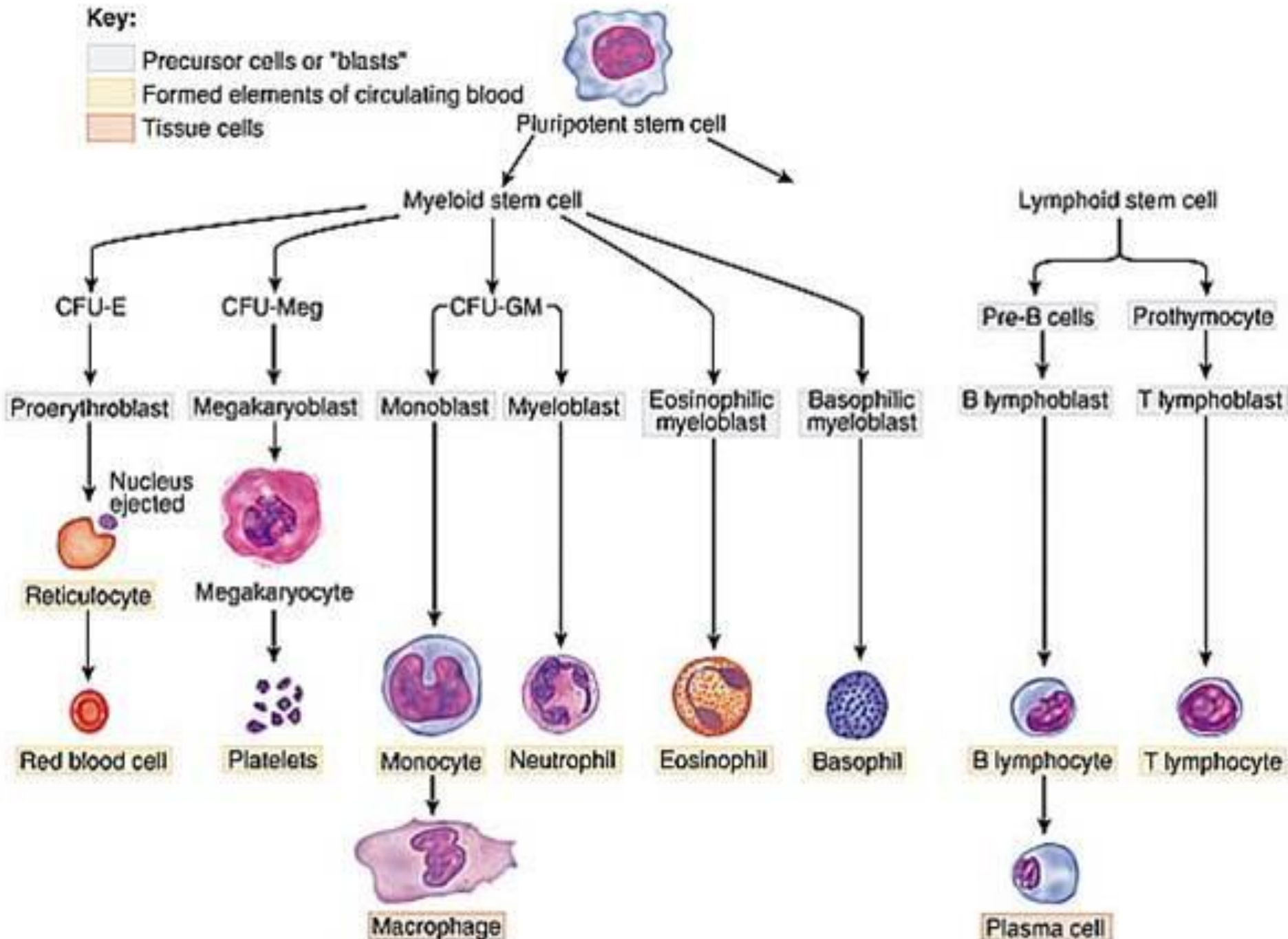
Cross-sectional view



Top view

Key:

-  Precursor cells or "blasts"
-  Formed elements of circulating blood
-  Tissue cells



PROCESS OF ERYTHROPOIESIS

- Stem cells are the primary cells capable of self-renewal and differentiating into specialized cells .
- Hemopoietic stem cells are the primitive cells in the bone marrow, which give rise to the blood cells & are called uncommitted pluripotent hemopoietic stem cells.
- PHSC is defined as a cell that can give rise to all types of blood cells. In early stages, the PHSC are not designed to form a particular type of blood cell. And it is also not possible to determine the blood cell to be developed from these cells: hence, the name uncommitted PHSC. In adults, only a few number of these cells are present. But the best source of these cells is the umbilical cord blood.

- When the cells are designed to form a particular type of blood cell, the uncommitted PHSCs are called committed PHSCs. Committed PHSC is defined as a cell, which is restricted to give rise to one group of blood cells.
- Committed PHSCs are of two types:
 1. **Lymphoid stem cells** (LSC) which give rise to lymphocytes and natural killer (NK) cells
 2. **Colony forming blastocytes**, which give rise to myeloid cells. Myeloid cells are the blood cells other than lymphocytes. When grown in cultures, these cells form colonies hence the name colony forming blastocytes.

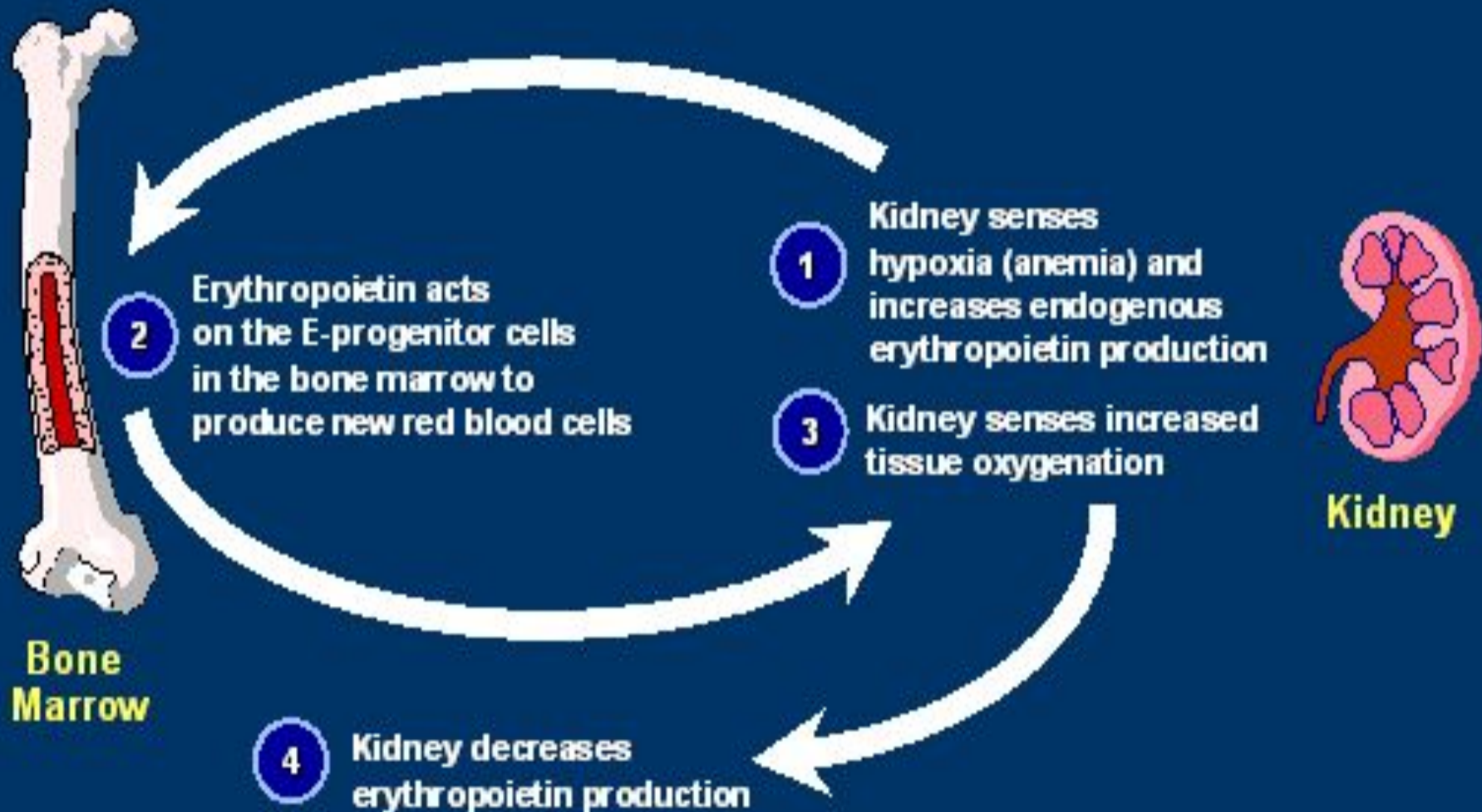
□ Different units of colony forming cells are:

i. Colony forming unit-erythrocytes (CFU-E) – Cells of this unit develop into erythrocytes

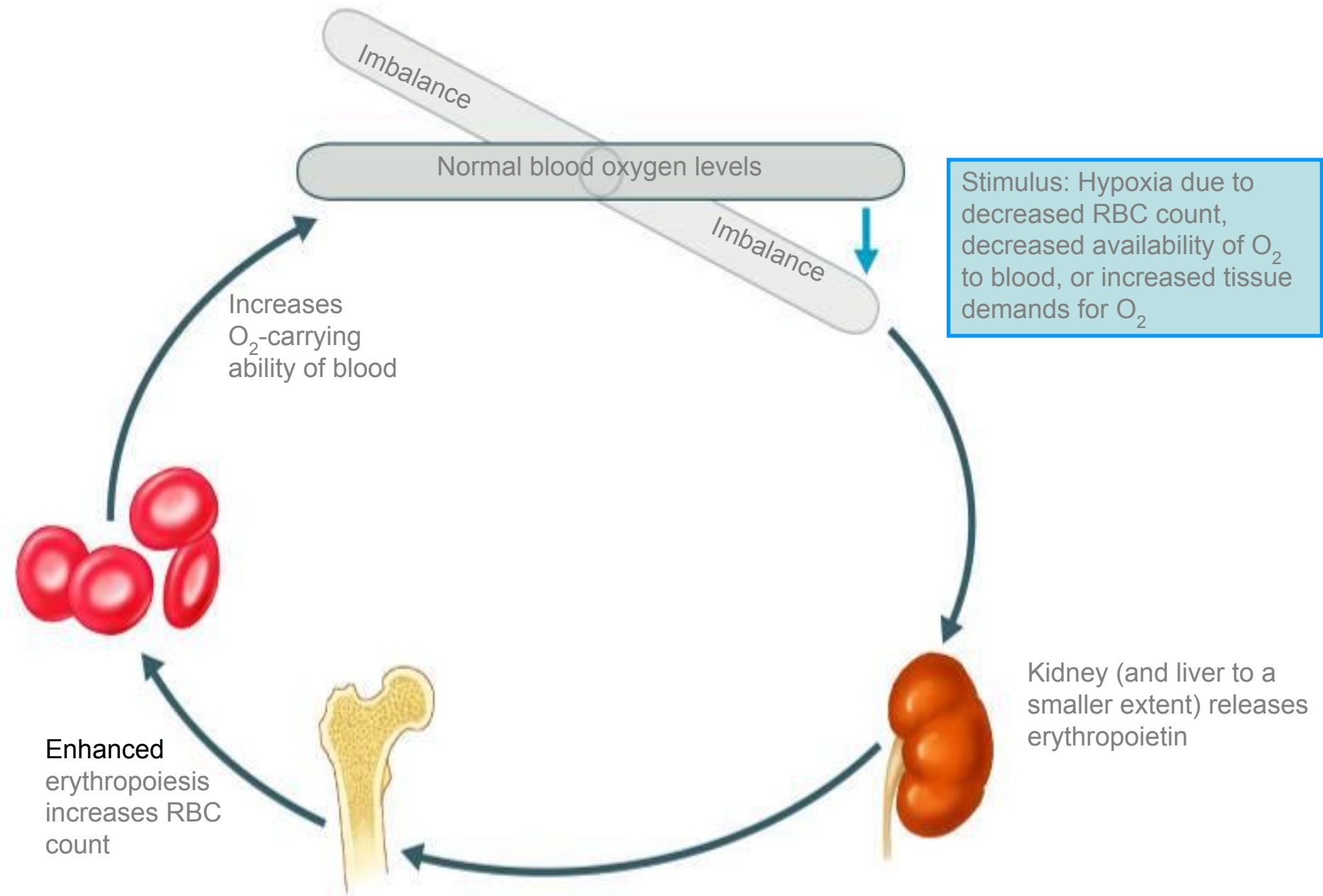
ii. Colony forming unit-granulocytes/monocytes (CFU-GM) – These cells give rise to granulocytes (neutrophils, basophils and eosinophils) and monocytes **iii.**

Colony forming unit-megakaryocytes (CFU-M) – Platelets are developed from these cells.

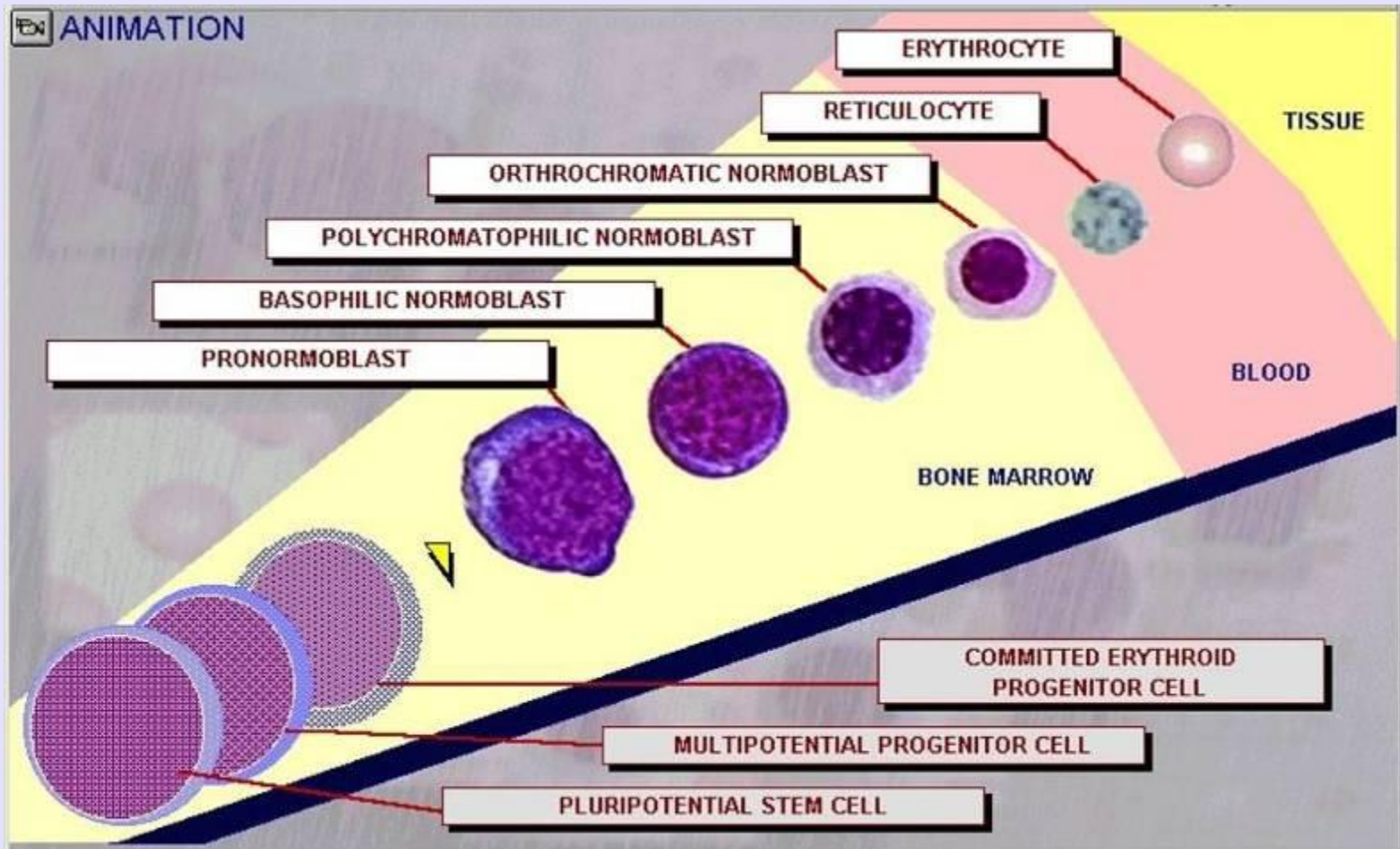
Erythropoiesis



ERYTHROPOIETIN MECHANISM



ERYTHROPOIESIS



CHANGES DURING ERYTHROPOIESIS

□ Cells of CFU-E pass through different stages and finally become the matured RBCs. During these stages four important changes are noticed.

1. Reduction in size of the cell (from the diameter of 25 to 7.2 μ)
2. Disappearance of nucleoli and nucleus
3. Appearance of hemoglobin
4. Change in the staining properties of the cytoplasm

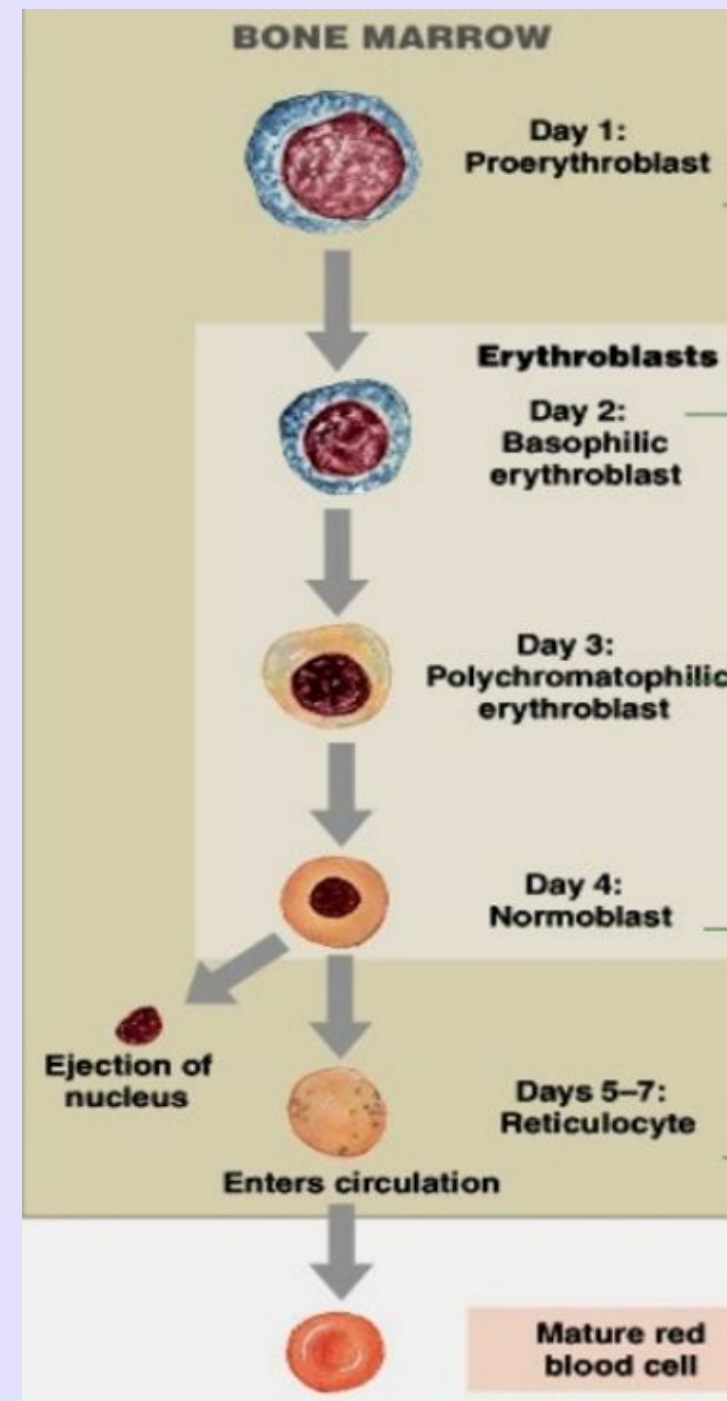
□ **STAGES OF ERYTHROPOIESIS**

□ Various stages between CFU-E cells and matured RBCs are :

1. Proerythroblast
2. Early normoblast (Basophil erythroblast)
3. Intermediate normoblast.(Polychromatophil erythroblast)
4. Late normoblast. (Orthochromatic erythroblast)
5. Reticulocyte
6. Matured erythrocyte.

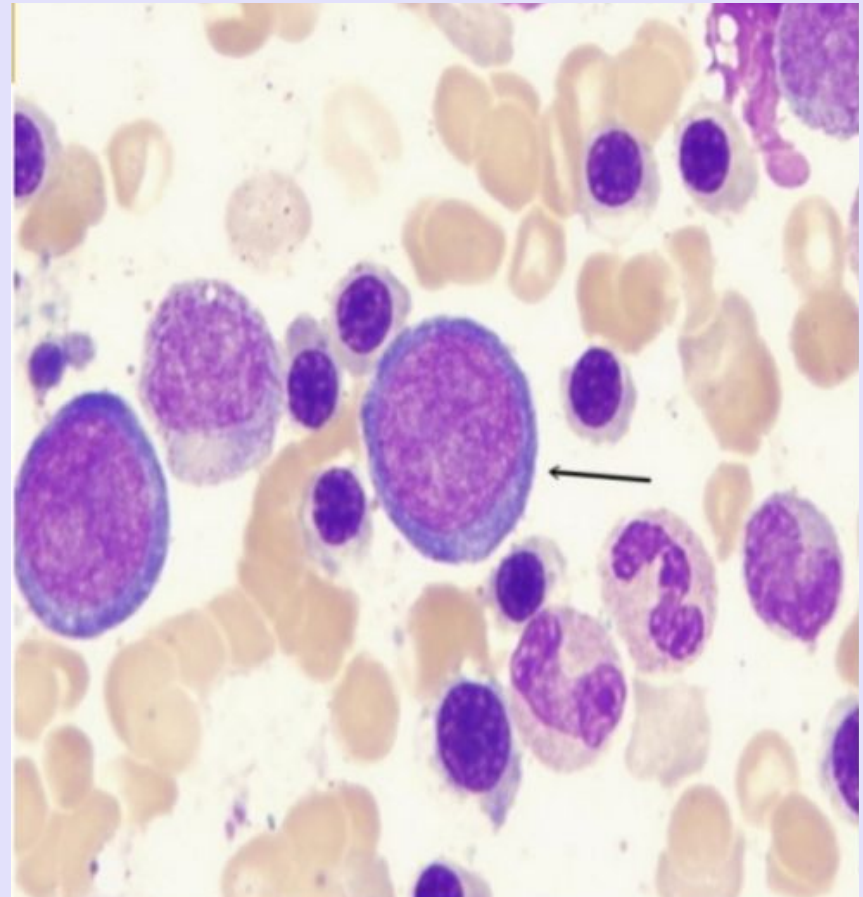
Erythropoiesis

- Occurs at a rate of **2-3 million RBC/sec**
- Maturation of erythroblast involves
 1. Decrease in **size** of cell
 2. Condensation and finally pyknosis of **nucleus**.
 3. Loss of **organelles** and machinery for protein synthesis.
 4. A change in **color** d/t accumulation of Hb.



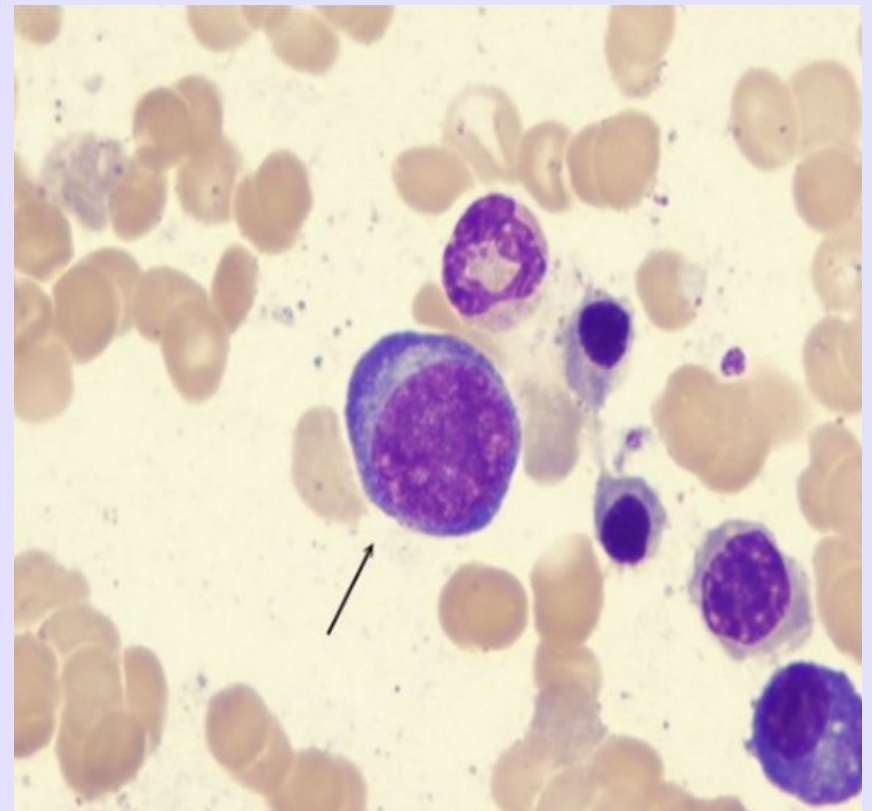
Proerythroblast

- **Size:** 15 to 20 μm
- High nuclear to cytoplasmic ratio
- Immature **nuclear chromatin**
- 1 – 2 **nucleoli**
- Basophilic **cytoplasm**
- No Hb
- Mitosis present
- Proerythroblast multiplies several times and finally forms the cell of next stage called early normoblast.



Basophil Erythroblast

- **Size:** 14 to 17 μm
- High nuclear to cytoplasmic ratio
- **Chromatin condenses**
- No nucleoli
- Active mitosis
- Basophilic **cytoplasm**
- **Hb synthesis** starts



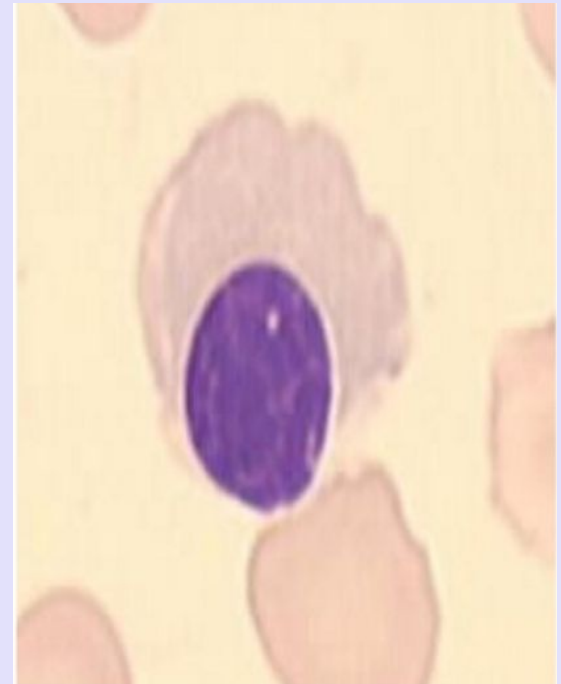
Polychromatophil Erythroblast

- ❑ **Size:** 10 to 15 μm
- ❑ Round nuc. with mature (**clumped chromatin**)
- ❑ No nucleoli
- ❑ **Hb appears**
- ❑ Cytoplasm is already basophilic. Now, because of the presence of hemoglobin, it stains with both acidic as well as basic stains. So this cell is called polychromophilic or polychromatic erythroblast
- ❑ Cell division minimal at this stage.



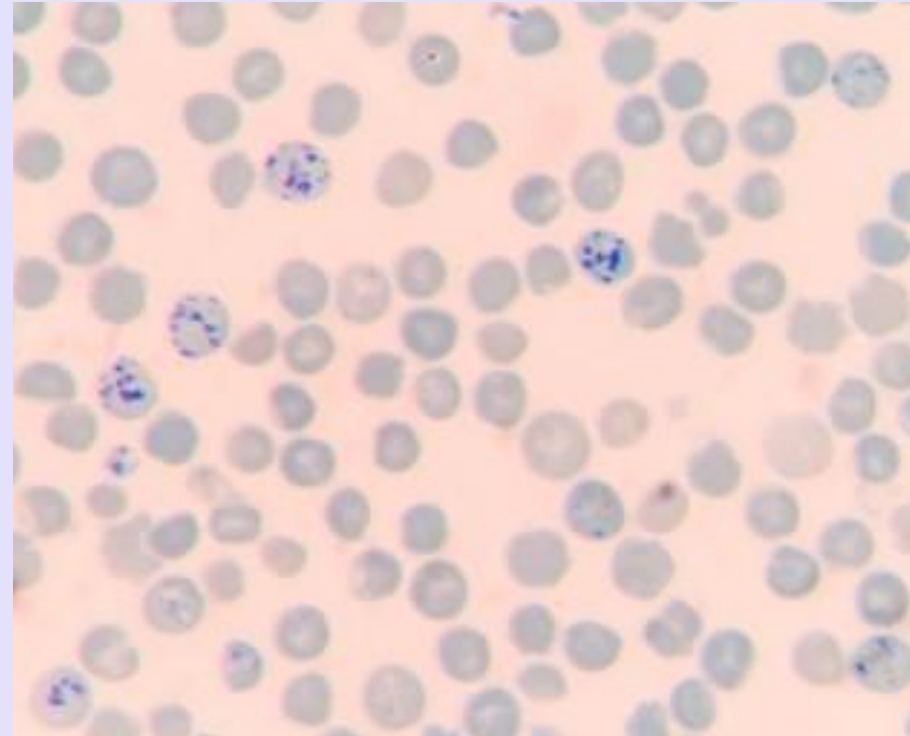
Orthochromatic Erythroblast

- ❑ **Size:** Diameter of the cell decreases further to about 8 to 10 μ .
- ❑ Nucleus becomes very small with very much condensed chromatin network and it is known as ink-spot nucleus.
- ❑ Quantity of hemoglobin increases. And the cytoplasm becomes almost acidophilic. So, the cell is now called orthochromic erythroblast.
- ❑ Mitosis absent
- ❑ In the final stage, just before it passes to next stage, the nucleus disintegrates and disappears. The process by which nucleus disappears is called pyknosis.



Reticulocyte

- Size: 7.8 - 8 μm
- Reticulocyte is otherwise known as immature RBC. It is slightly larger than matured RBC
- No nucleus, no organelle
- The cytoplasm contains the reticular network or reticulum, which is formed by remnants of disintegrated organelles. Due to the reticular network, the cell is called reticulocyte.
- Final 20% Hb formed
- 0.5 to 2 % of all RBCs
- Reticulocytosis seen following hemolysis or acute blood loss
- ↓ Reticulocyte count in Aplastic anemia

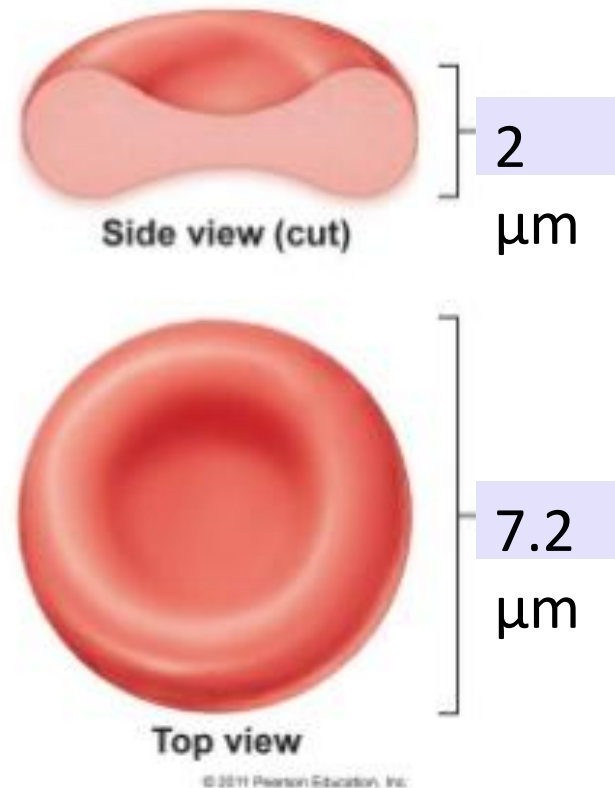


Matured Erythrocyte

- Reticular network disappears and the cell becomes the matured RBC and attains the biconcave shape.
- The cell decreases in size to $7.2\ \mu$ diameter.
- The matured RBC is with hemoglobin but without nucleus.
- It requires 7 days for the development and maturation of RBC from proerythroblast. It requires 5 days up to the stage of reticulocyte. Reticulocyte takes 2 more days to become the matured RBC.

Erythrocytes (RBCs)

- Biconcave disc
 - Large surface area : volume ratio
- No nucleus
- No organelles
- 120 day life span
- Filled with **hemoglobin (Hb)**
 - transports O_2

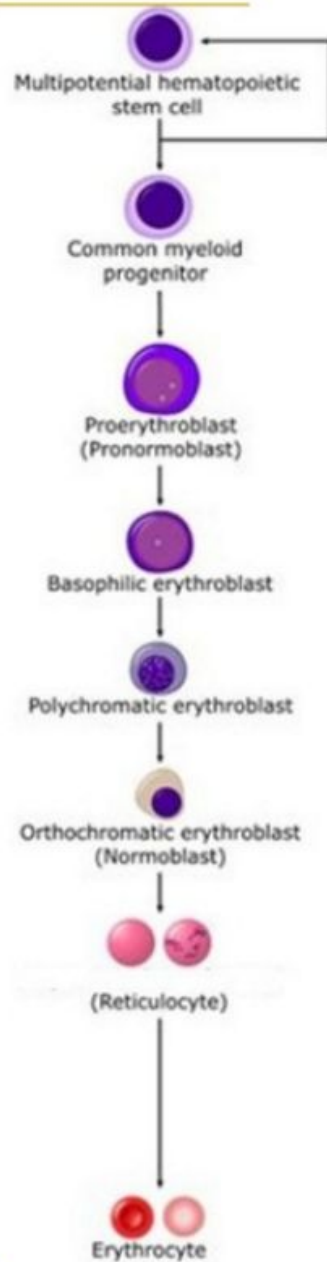


Duration of erythropoiesis

HSC to RBC- 21 days

Differentiation phase: from
pronormoblast to
reticulocyte phase- 5 days

Maturation phase: from
reticulocyte to RBC- 2 days



7

Factors Affecting Erythropoiesis

FACTORS NECESSARY FOR ERYTHROPOIESIS

- Development and maturation of erythrocytes require variety of factors, which are classified into three categories:
 1. General factors
 2. Maturation factors
 3. Factors necessary for hemoglobin formation

FACTORS

□ **GENERAL FACTORS** necessary for erythropoiesis are:

- i. Erythropoietin
- ii. Thyroxine
- iii. Hemopoietic growth factors
- iv. Vitamins.

□ **MATURATION FACTORS**

- i. Vitamin B12,
- ii. intrinsic factor and
- iii. folic acid are necessary for the maturation of RBC

ERYTHROPOIETIN

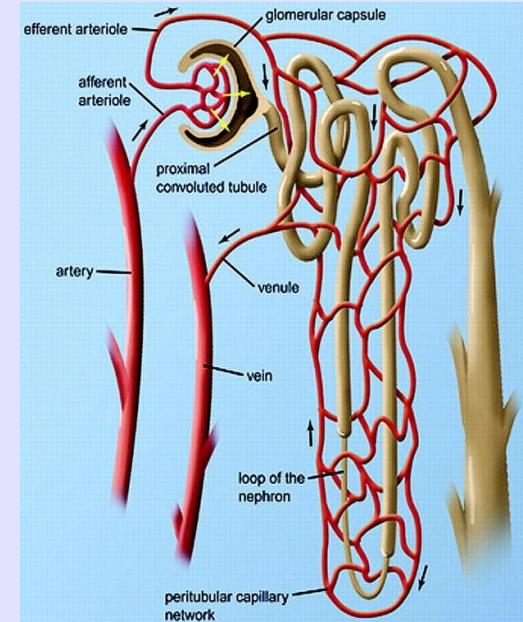
- Glycoprotein MW-34000 (165 AA residues)

Formation

- 85% formed in endothelial cells of the peritubular capillaries of the renal tubules.
- 15% formed in liver, hepatic cells & Kupffer cells.

Breakdown

- In liver. Half life is 5hours



ERYTHROPOEITIN

Stimuli for production

- ❖ **Hypoxia**
- ❖ Products of RBC destruction
- ❖ High altitude
- ❖ Anemia
- ❖ Chronic lung or kidney diseases
- ❖ Catecholamines
- ❖ Prostaglandins

Androgens

Inhibition

- ❖ Blood transfusion

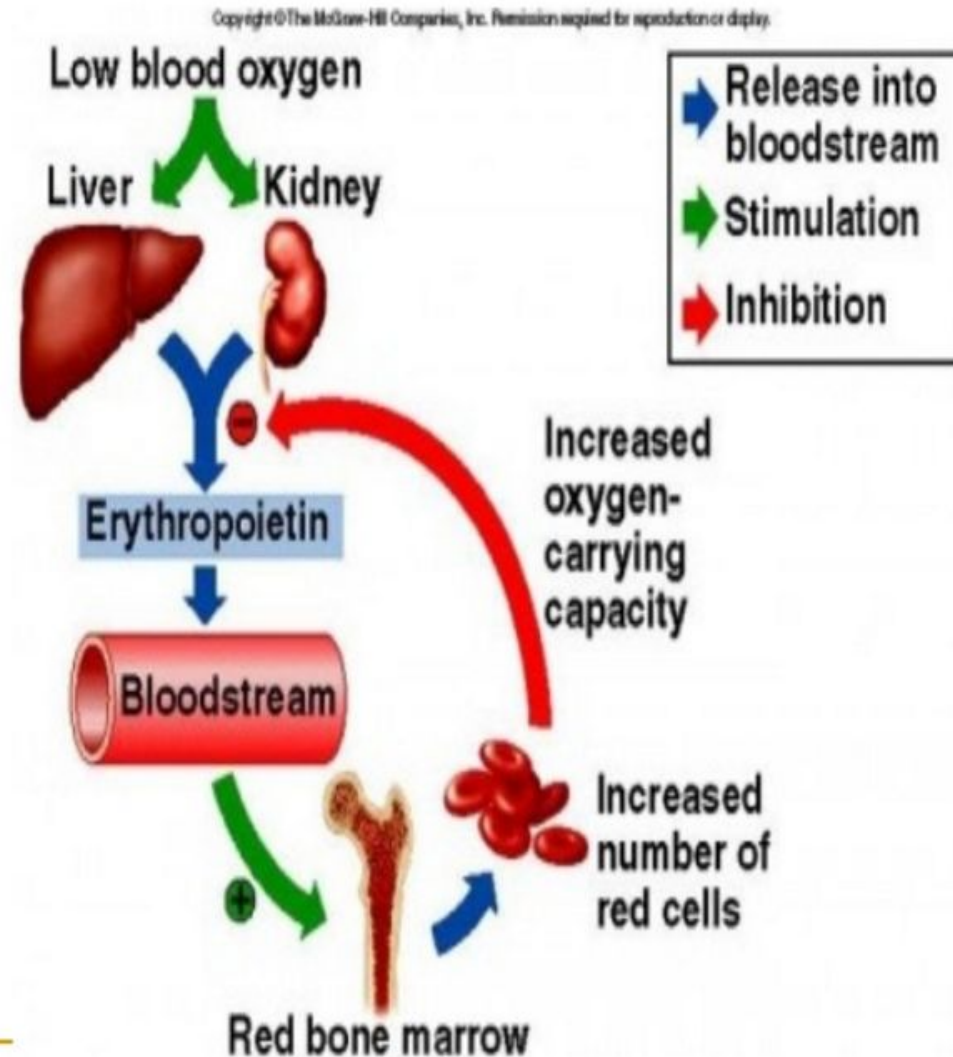


Functions of Erythropoietin

Erythropoietin increases RBC

production in 3 ways:

- ❑ Promotes pronormoblast production
- ❑ Shortens the transition time through the normoblast stage
- ❑ Promotes the early release of reticulocytes.



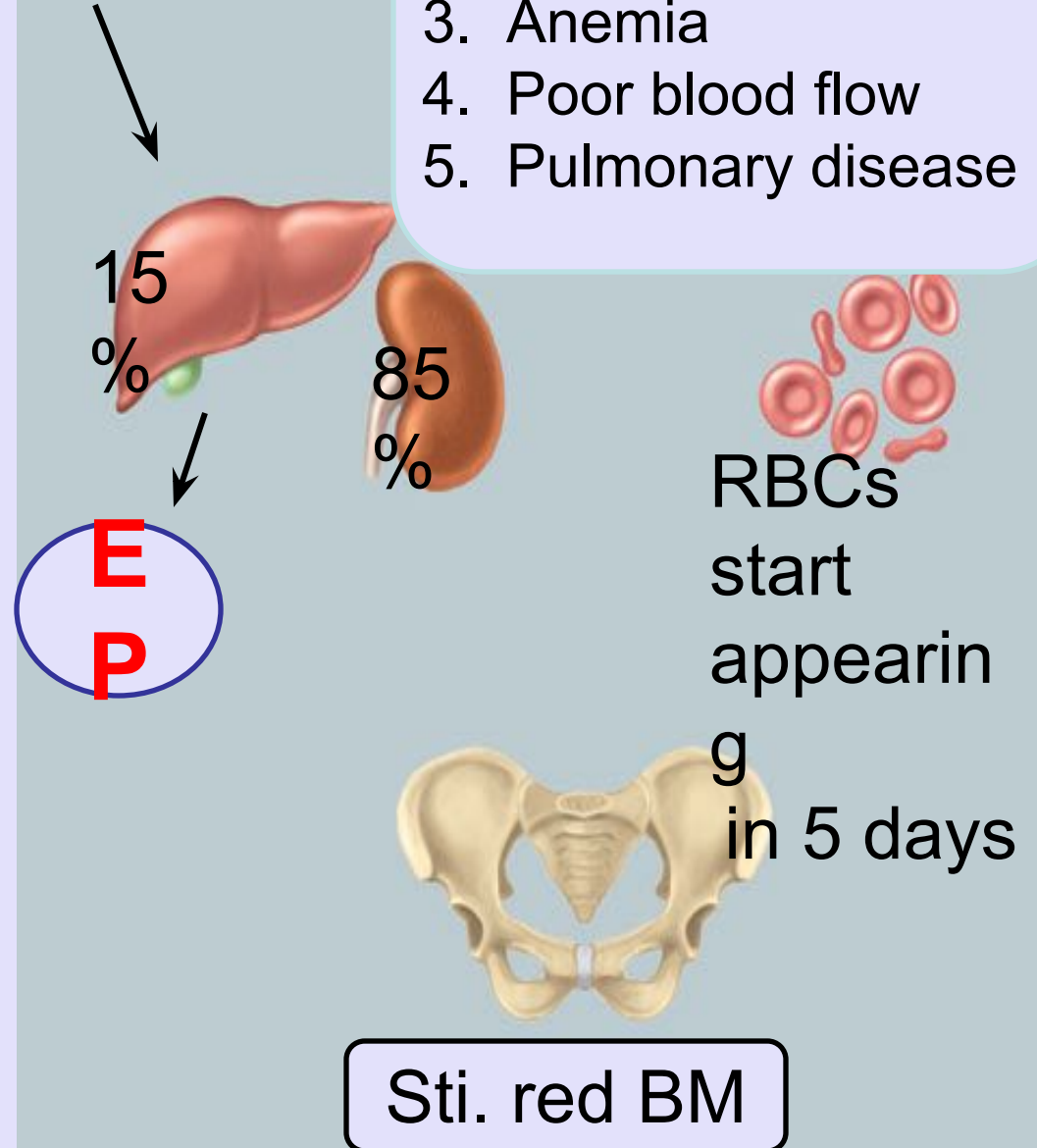
Erythropoietin

- **Mode of Action**

- ↑ No. of Stem cells & CFU-E
- ↑ Hb synthesis in precursor cells
- Influences the rate of maturation of precursor cells
- Controls release of RBCs from marrow into circulation

↓ Tissue Oxygenation

1. Low blood volume
2. Low Hb
3. Anemia
4. Poor blood flow
5. Pulmonary disease



Hemopoietic Growth Factors

- Hemopoietic growth factors or growth inducers are the interleukins and stem cell factor (steel factor). Generally these factors induce the proliferation of PHSCs. Interleukins (IL) are glycoproteins, which belong to the cytokines family.
- Interleukins involved in erythropoiesis:
 - a. Interleukin-3 (IL-3) secreted by T-cells
 - b. Interleukin-6 (IL-6) secreted by T-cells, endothelial cells and macrophages
 - c. Interleukin-11 (IL-11) secreted by osteoblast.

■ General Factors

- Hypoxia □ Erythropoietin
- Growth inducers: e.g. Interleukins (esp. IL3)
- Vitamins

■ Maturation Factors

- Vitamin B12 & Intrinsic factor
- Folic acid

■ Factors necessary for Hb production

- Vitamin C □ Helps in iron absorption (Fe^{3+} □ Fe^{2+})
- Proteins □ Amino Acids for Globin synthesis
- Iron & Copper □ Heme synthesis
- Calcium, Bile salts, Cobalt & Nickel

THANK
YOU!
